Text Document

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Machine Learning Model Design
Overview
The Flood Prediction System uses machine learning algorithms to identify patterns in historical flood data and predict future flood risks.

Machine Learning Workflow
1. Data Collection
2. Data Preprocessing
3. Feature Selection
4. Model Training
5. Model Evaluation
6. Model Deployment

Input Features
The model uses the following features:
- Rainfall
- Latitude
- Longitude
- Elevation
- Slope
- Distance from water bodies
- Flood duration

Algorithms Used
Decision Tree
Decision Tree creates a tree-based decision structure to classify flood risks based on input parameters.

Random Forest

Random Forest combines multiple decision trees to improve prediction accuracy and reduce errors.

K-Nearest Neighbor (KNN)
KNN predicts flood risk by comparing new data points with similar historical data.

XGBoost
XGBoost is an advanced boosting algorithm that improves prediction performance by combining multiple weak models.

Model Evaluation Metrics
The model performance is evaluated using:
- Accuracy
- Precision
- Recall
- F1 Score